SOC Detection Validation & Alert Investigation

Case ID: SOC-LAB-2026-001
Case Title: Windows Failed Authentication — Below-Threshold Validation
Case Type: Detection Rule Validation / Security Event Review
Analyst: Bibek Kunwar
Environment: Windows Virtual Machine

Status: Pending Evidence Verification
Severity: Informational — Controlled Lab Test
Detection Rule: Five Failed Logons Within Five Minutes

1. Executive Summary

The Windows authentication check using Python was tested on the EVTX security logs that were extracted. The test looks for Windows Security Event ID 4625 and gives an alert in case of the fifth failed logon for any user and source IP address combination within the five minutes time span. In the screenshots provided below, there is no alert for a one event test run with the standard threshold, no alert for four events, an alert for a one event test run with a decreased threshold, an alert for eight events, and no alert for six events. The latter requires time stamp validation to confirm success.

2. Scope and Environment

Item	Description
Data source	Windows Security EVTX exports
Analysis workstation	Ubuntu Linux (terminal screenshots)
Test endpoint	Windows virtual machine
Tool	Python 3, python-evtx, ElementTree, datetime, deque
Detection rule	5 failures / 5 minutes / same target username + source IP
Event ID	4625 (failed logon)
Analysis mode	Offline processing of exported EVTX files

3.Detection Login and Implementation

In the uploaded detect_bruteforce.py script, LOG_FILE is set to "final_5more_failed.evtx", while the values of the constants

THRESHOLD and WINDOW_MINUTES are 5 and 5, respectively. The script extracts each record from the log file as XML and keeps only those events whose EventID is 4625. Then, it extracts SystemTime, TargetUserName, and IpAddress, sorts the events by timestamp, and keeps a list of the most recent timestamps for each (username, IP) combination, which are older than current_time - 5 minutes. The script outputs an alert message whenever the window length becomes 5.

4. Test matrix and observed results

ID	Scenario	Observed output	Assessment
T01	1 failure; rule 5 in 5 min	1 event; no alert	Expected negative result
T02	1 failure; rule 1 in 1 min	1 event; 1 alert	Threshold sensitivity demonstrated; not normal production rule
T03	4 failures; rule 5 in 5 min	4 events; no alert	Expected below-threshold result
T04	6 events; rule 5 in 5 min	6 events; no alert	No threshold reached in script; per-event times/grouping not supplied

T05	8 events; rule 5 in 5 min	8 events; 1 alert	Positive detection demonstrated

5. Positive alert investigation — T05

Field	Observed value
Account	IEUser
Source IP	127.0.0.1
Failure count reported at trigger	5
Rule window	5 minutes
Trigger event (UTC)	2026-10-22T15:57:17.337746+00:00
Total matching events in file	8
Alerts generated	1

The script output supports that the configured correlation threshold was met in the processed data. The IP address 127.0.0.1 is loopback, not evidence of an external network attacker. No account takeover, successful login, or malicious actor has been established from these screenshots.

6. Negative test analysis — T03 and T04

T03: Four failed logon events were counted while the rule required five in five minutes. No alert was generated. This is the expected below-threshold behavior. The screenshot does not independently display each timestamp or source field; it confirms the script count and disposition.

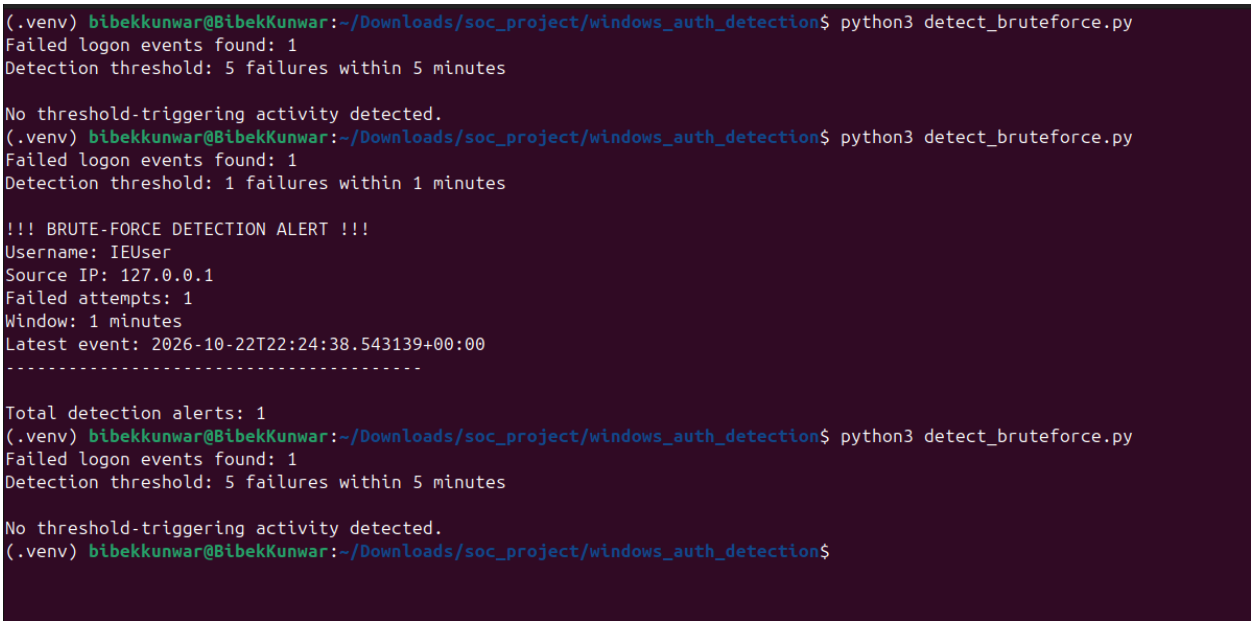
T04: Six failed events were counted, but no alert was generated. This does not contradict the rule: a total of six in an EVTX file is not equivalent to five matching events in one rolling five-minute group.

7. Evidence Registers

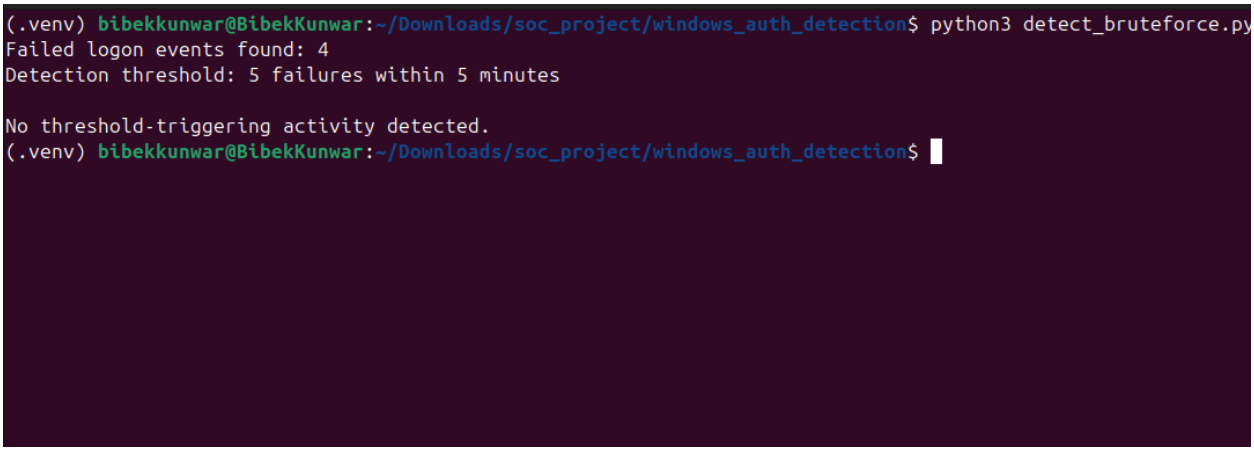
Ref	Evidence	What it supports
E01	one and multiple failure event.png	One-event normal threshold; altered 1-in-1-minute alert; reset to normal rule
E02	4_failed_event.png	Four-event count with no alert
E03	6_failedacroess5minutes.png	Six-event count with no alert
E04	5_more_failed_event.png	Eight-event positive alert with IEUser and 127.0.0.1
E05	detect_bruteforce.py	Actual parsing, grouping and alert implementation

8. Evidence screenshots

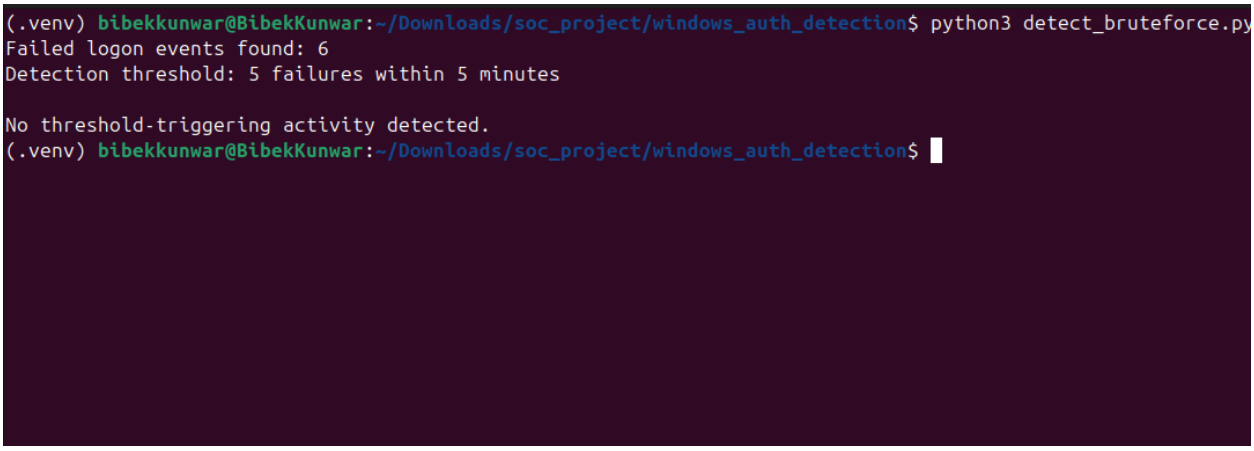
E01 — One-event and sensitivity tests



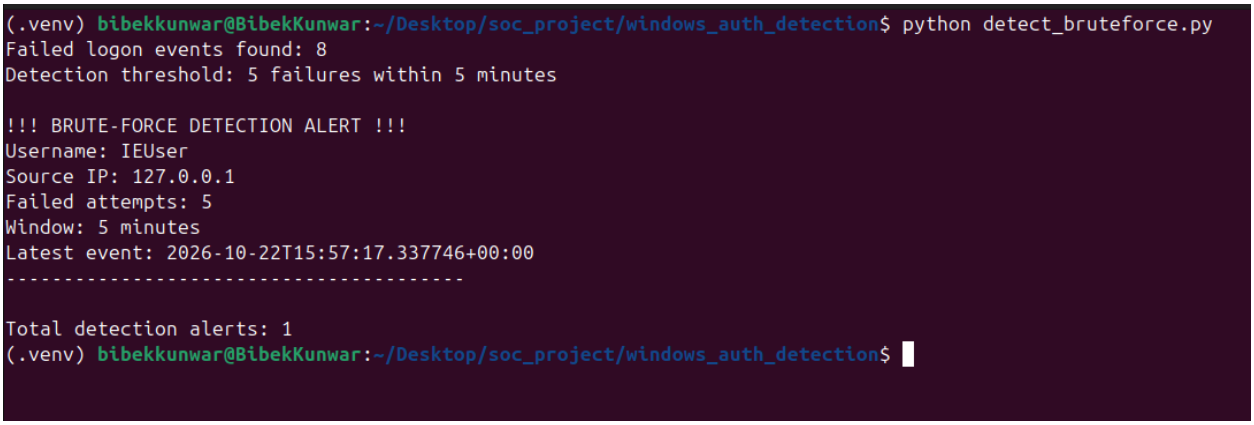
E02 — Four-event negative test



E03 — Six-event no-alert test



E04 — Eight-event positive detection



9. SOC Disposition and Recommended Next Actions

Disposition: Controlled laboratory detection validation; no confirmed security incident. T01, T02, T03, and T05 outputs align with their configured thresholds. T04 is an observed no-alert result whose exact

cause should be validated from event-level fields. No escalation is indicated solely on the basis of these authorized test screenshots.

Next actions:

1. Capture event-level timestamps, usernames, and source IP addresses for Test Case T04.
2. Preserve the original EVTX files and record their SHA-256 hashes and collection timestamps to maintain evidence of integrity.
3. Document the exact EVTX input file used for each detection test.
4. Include additional Windows Event Log fields, such as LogonType and authentication failure codes, to improve SOC investigations.
5. Implement automated regression tests to validate detection of accuracy and export alerts in JSON or CSV format.

10. Limitations

The screenshots prove terminal outputs, not independently of the provenance, completeness, or exact timing of every underlying event. The uploaded Python file shows the current configuration, while the screenshots show different input datasets and one deliberately modified threshold. This is an offline proof-of-concept detector, not a deployed

SIEM alert rule. No incident severity beyond informational lab testing is asserted.

11. Five Ws and One H — alert triage

Who: The targeted Windows account is IEUser. The actor or process responsible for the failed logons has not been established.

What: The positive test input contained eight Event ID 4625 records; five qualifying failures triggered one alert.

When: The threshold-triggering event was logged at 2026-10-22 15:57:17.337746+00:00. The screenshot does not provide all preceding timestamps.

Where: The events originated from the Windows virtual-machine Security log. The reported source IP was 127.0.0.1, a loopback address, not an external source.

Why: The alert fired because the configured correlation rule reached five failures for one username/source-IP combination within five minutes.

How: Python reads the exported EVTX records, filtered Event ID 4625, extracted timestamps/user/IP, sorted events, and applied a rolling five-minute deque per username/IP pair.

12. Conclusion

The evidence demonstrates a working prototype for identifying repeated Windows failed-logon events using a username/IP-correlated rolling five-minute rule. The positive eight-event test produced an alert, and the four-event negative test did not. This is suitable as a junior SOC portfolio case study when presented as a detection validation exercise,

with the six-event timing explanation explicitly marked pending verification.

Reference

Microsoft (n.d.) 4625(F): An account failed to log on. Microsoft Learn. Available at: https://learn.microsoft.com/en-us/windows/security/threat-protection/auditing/event-4625